JGI JAIN DEEMED-TO-BE UNIVERSITY

आविष्कार FET HACKATHON

FET Alumni and Outreach
CONNECT INNOVATE IMPACT

AVISHKAR 2026 · AI & IoT TRACK

MixCheck

Same coding task. Different language.
Does the AI still get it right?

A robustness tester for AI coding assistants, using prompts in code-mixed Indian languages.

</> Cross-lingual code-switching robustness tester

Try it live →

Team El Rata Alada — Tanmay, Ritobroto, Aaditya, Aaliya

HINGLISH

yeh function crash **kyu ho raha hai?**

BENGLISH

ei list **ta** sort **kore dao**

KANGLISH

bari code **kodu**

TANGLISH

indha code-la enna error?

ODIA-ENGLISH

ei function **ta** fix **karida**

● Live: mixcheck-elrata.streamlit.app

Problem Statement

AI coding assistants are tested in clean English. Indians ask them in mixed languages.

314.9M

Indians are bilingual

Census of India, 2011

Many think and type in mixed languages, including when they ask AI to write code.

Tested in English

HumanEval-XL covers 23 languages. None is Indian, none mixed.

Nobody measures the gap

Students get broken code from mixed prompts and never know why.

SPOT THE FAILURE · A REAL ANSWER FROM OUR TESTS

casual Benglish → GPT-OSS 120B

THE PROMPT

bhai python **e** `is_palindrome` **naam er** fn **bana,** string **e kono** space **thakbe na, shamne theke aar pichon theke** same **porle** True **noile** False, capital small **er tofat korbi na. shudhu** code

the AI's reply

```
def is_palindrome(s: str) -> bool:
    return s == s[::-1]
```

5 hidden tests, e.g. is_palindrome("Racecar") should be True

Will this code pass all 5 tests?

Click to answer

✓ It passes

× It fails

Solution

Same coding task in English and in mixed Indian languages. MixCheck runs the code and measures the gap.

1. Paired prompts

25 tasks, each in 10 versions. Only the language changes.

2. Run on 3 AI models

Gemini, GPT-OSS 120B and GPT-OSS 20B.

3. Two scores

Does the code pass every test? Did it reply with only the code, as asked?

Open the live dashboard →

mixcheck-elrata.streamlit.app

THE MIX DIAL · CLICK A LANGUAGE

Task: `indian_format(1234567) → "12,34,567"`

English | Hinglish | Benglish | Kanglish

Write a Python function named `indian_format` that takes a non-negative whole number and returns it as a string with commas in the Indian style: the last three digits form one group, and every two digits before that form a group. Numbers below `1000` have no commas. Example: `indian_format(1234567)` returns `"12,34,567"`. Reply with only the code.

0% of words not in English

Gemini	GPT-OSS 120B	GPT-OSS 20B
Passed 6 of 6	Passed 6 of 6	Passed 6 of 6

Clean English: all three models pass.

Tech Stack + Workflow

From prompt to verdict, fully automatic, with untrusted AI code kept in a sandbox.



Data

- Python 3
- pandas
- 250 prompts
- 25 test files

Models

- Gemini API
- Groq API
- GPT-OSS 120B / 20B
- temperature 0

Grading

- separate process
- 10 s time limit
- audit-hook sandbox
- no API keys inside

Dashboard

- Streamlit
- Plotly
- custom HTML/JS widgets
- Streamlit Cloud

48 automated checks · a reference solution for every task · **deployed live** · free API tiers, no GPU

Innovation

Existing benchmarks don't test code written from code-mixed Indian prompts. MixCheck does.

Paired prompts

Same task, only the language changes, so any drop comes from the mixing.

Graded by running code

Test cases decide. No opinions, no hand-labelling.

Two scores

Does the code work? Did the AI follow the instructions?

Statistics built in

95% ranges and Fisher's exact test separate real gaps from chance.

	Indian languages	Mixed prompts	Tests code	Checks instructions
HumanEval-XL	✗	✗	✓	✗
LinCE	✓	✓	✗	✗
GLUECoS	✓	✓	✗	✗
MixCheck	✓	✓	✓	✓

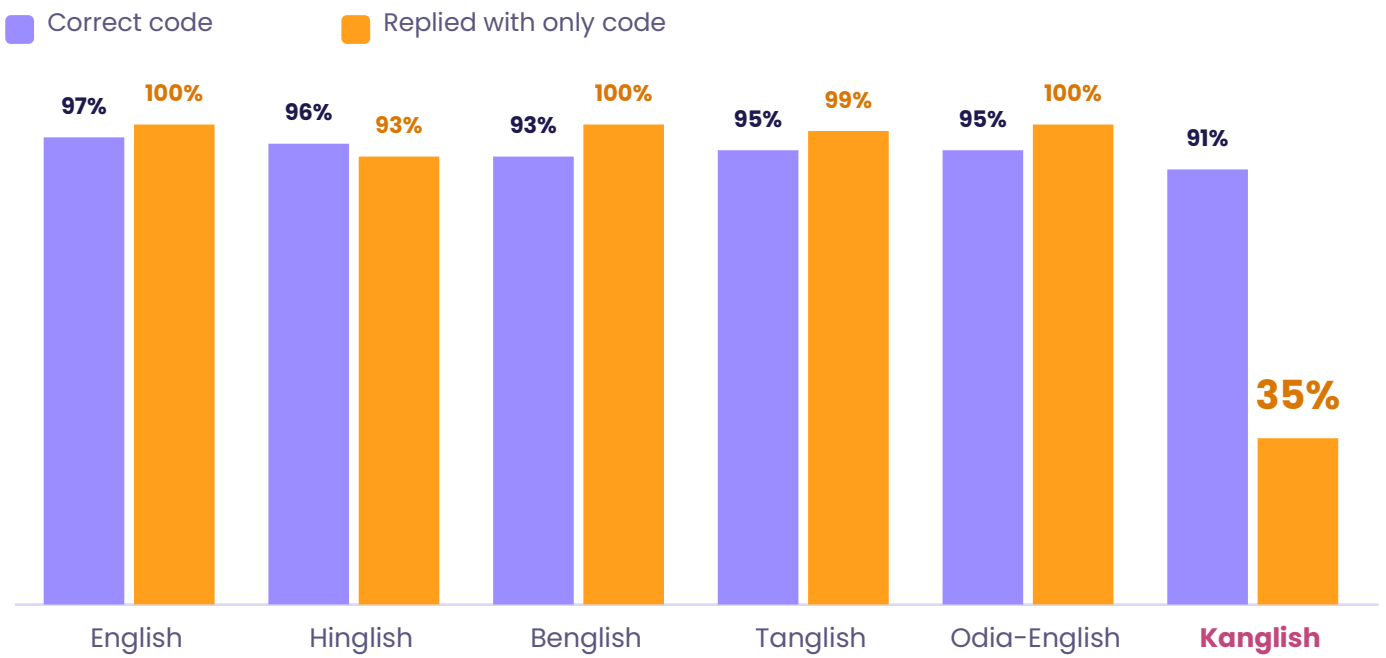
LinCE and GLUECoS include Hindi-English but test language understanding, not code. HumanEval-XL has 23 languages, none Indian.

Impact

What we measured across 750 graded answers: the code mostly works, the instructions break.

Same casual style, only the language changes

3 models × 25 tasks per language · one run each · temperature 0



35% vs 100%

Kanglish answers that replied with only the code, vs casual English. $p < 0.001$: not chance.

9 vs 0

Renamed functions: 600 mixed prompts vs 150 English ones. One became bhai_indian_format.

95.3%

of all 750 answers had correct code. The gaps in correct code are small and could be chance.

Target Audience / Market

Everyone building, teaching with or using AI coding tools in India.

AI tool builders

Find language blind spots before an Indian launch.

Colleges and edtech

Students ask in Hinglish. Know when answers can be trusted.

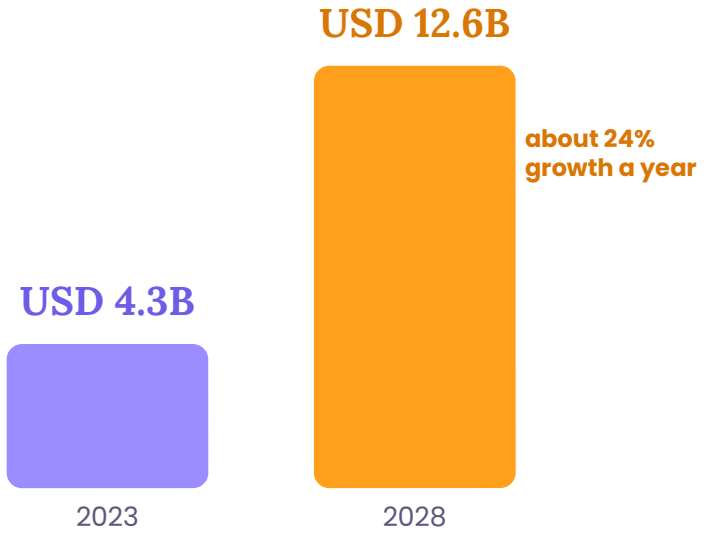
Researchers

An open benchmark: a new language is one new column.

Indian developers

Asking in your own language shouldn't cost working code.

AI CODE TOOLS MARKET



Source: MarketsandMarkets

314.9M bilingual Indians who could be typing those prompts (Census 2011).

Revenue Model

A free open core builds trust. Teams pay to test their own models.

Open-source core

Free

- Benchmark + dashboard on GitHub
- Community adds languages
- Builds trust and adoption

For students and researchers

MOST TEAMS

MixCheck Pro

₹799 / month

- Test your own model via an API
- Private reports for your team
- Alerts when an update gets worse

For startups and product teams

Custom audits

One-time fee

- Pre-launch testing of your product
- Your languages and domains
- A “MixCheck Verified” report

For companies launching in India

Future Improvements

From a benchmark to a safety net for every AI coding tool used in India.

NOW

Started today

A native Kannada speaker reviewed our Kanglish prompts. A fix experiment tests an English guard line vs translate-first.

NEXT

Wider and stronger

Native review of every prompt. Marathi, Telugu, Malayalam. Several runs per prompt.

THEN

Fix it, not just find it

A plug-in for coding tools that applies the fix that works best, plus voice prompts.

VISION

Public leaderboard

Re-run MixCheck on every model update and publish the results for everyone.

Goal: AI coding tools that work as well in Kanglish as they do in English.

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MixCheck

Thank You

Questions?



Scan or click to try MixCheck live

mixcheck-elrata.streamlit.app

T — Tanmay
Team lead & developer

Builds the app, runs the live demo

R — Ritobroto
Testing & evaluation

Runs the models, checks the results

A — Aaditya
Dataset & design

Coding tasks, prompts, design

A — Aaliya
Research & presentation

Problem research, the pitch